

Figure 7: Flow graph generation for the analysed packets

is classically used as a passive network sniffer/packet capturing tool to detect the operating systems, sessions, hostnames, open ports and related information without placing any traffic on the network.

Using NetworkMiner

The GUI of NetworkMiner is divided into tabs. Each tab has a different approach towards analysing information of the captured data.

The followings steps are used to analyse network traffic. First, select the network interface for which the data has to be captured.

By default, the *Hosts* tab is selected. You can sort hosts on the basis of IP address, MAC address, hostname, operating system, etc.

Press the *Start* button to begin the sniffing process.

Snort

Snort is an open source tool written in C, used as a network intrusion prevention and detection system (IDS/IPS) and has been developed by Sourcefire. It offers an excellent combination of benefits like signature, protocol and anomaly-based inspection. With millions of downloads and nearly 400,000 registered users, Snort has become the de facto standard for IPS.

Snort can implement protocol analysis and content investigation with a number of other features including detection of a variety of attacks and probes such as buffer overflows, stealth port scans, CGI attacks, SMB probes, OS fingerprinting attempts, and more. Snort makes use of a flexible rules language to explain the traffic that it should collect or pass, as well as a detection engine that has a

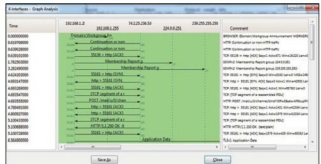


Figure 9: Analysis of all packets

modular plug-in architecture.

Snort can be configured in three different modes: sniffer, packet logger and network intrusion detection. In sniffer mode, the tool reads the network packets and displays them on the console. In packet logger mode, the tool implements the logging of the packets to the disk. In intrusion detection mode, the tool monitors the network traffic and analyses it against a rule set defined by the user.

The main configuration file is */etc/snort/snort.conf*. In this configuration file, the actual information of the network or system that is under investigation is specified. All values and parameters are commented in the file so that the changes can be done very easily.

Some of the extracts from the configuration file are:

```
# Setup the network addresses you are protecting
ipvar HOME_NET any

# Set up the external network addresses. Leave as "any" in
most situations
ipvar EXTERNAL_NET any

# List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET

# List of SMTP servers on your network
ipvar SMTP_SERVERS $HOME_NET

# List of web servers on your network
ipvar HTTP_SERVERS $HOME_NET

# List of sql servers on your network
ipvar SQL_SERVERS $HOME_NET

# List of telnet servers on your network
ipvar TELNET_SERVERS $HOME_NET

# List of ssh servers on your network
ipvar SSH_SERVERS $HOME_NET

# List of ftp servers on your network
ipvar FTP_SERVERS $HOME_NET

# List of sip servers on your network
ipvar SIP_SERVERS $HOME_NET

# List of ports you run web servers on
portvar HTTP_PORTS {36,80,81,82,83,84,85,86,87,88,89,90,311,3
83,555,591,593,631,801,808,818,901,972,1158,1220,1414,1533,17
41,1830,2231,2381,2381,2809,3029,3037,3057,3128,3443,3702,400
0,4343,4848,5117,5250,6080,6173,6988,7060,7061,7144,7145,7510
,7770,7777,7779,8060,8068,8014,8028,8080,8081,8082,8085,8088,
```